BRAC Bank POC Implementation Plan

Requirement Analysis + Solution Design Document

1. Executive Summary

Comptech will deliver a controlled Proof of Concept to demonstrate implementation capability across OpenShift, observability, identity and API management, CI/CD, security scanning, high-availability data services, event streaming, middleware, and enterprise Java application hosting.

The POC will be executed in a **vendor-hosted internal lab environment** using:

- **Two 3-node OpenShift clusters** provisioned on VMs using Terraform:
  - **DC cluster**
  - **DR cluster**
- **GitOps-driven OpenShift delivery** using Argo CD with an **agent-based pull model**
- **VM-hosted enterprise shared platforms** for WSO2, GitLab, Jenkins, Nexus, Vault, Splunk, PostgreSQL, and JBoss
- **OpenShift-hosted application and platform services** for B-TOP, Redis, Kafka, observability, logging, and supporting operators
- **Vault-based secret management** and **Vault PKI via cert-manager** for internal certificates
- **Public CA / Let's Encrypt via cert-manager** for external browser-facing endpoints
- A realistic banking demo application architecture built around:
  - **B-TOP**: Bank Transaction Operations Platform on Open Liberty
  - **B-ORP**: Bank Operations Reconciliation Portal on JBoss

The POC will demonstrate:

- infrastructure automation
- operator-driven platform installation
- application delivery through GitOps
- end-to-end observability
- security enforcement
- auditability

- RBAC
- rate limiting
- Redis HA failover
- Kafka schema enforcement and DLQ
- logging to external SIEM-style tooling
- DR readiness and controlled failover approach

2. POC Objectives

The objective of this POC is to prove Comptech's ability to:

1. provision and operate OpenShift through infrastructure-as-code
2. deploy and secure containerized workloads
3. implement modern observability for logs, metrics, and traces
4. deploy and configure enterprise middleware and API management
5. automate CI/CD and GitOps workflows
6. enforce image and dependency security controls
7. demonstrate HA and DR patterns across core services
8. present a realistic banking-oriented application landscape, rather than isolated technical demos

3. Scope Summary

The BRAC Bank mail requests implementation and justification across the following areas: OpenShift Platform, Logging & Observability, WSO2, CI/CD & DevOps tooling, Trivy, Redis HA, Kafka, Open Liberty + NGINX, and JBoss Domain Mode.

This POC addresses all of those areas through a single integrated design.

4. Final Solution Overview

4.1 Platform split

On OpenShift

- B-TOP on Open Liberty
- Redis HA
- Kafka on Strimzi in KRaft mode
- Schema Registry
- Kafka Connect
- OpenTelemetry collectors
- SigNoz
- ClickHouse
- OpenShift operators and cluster services
- cert-manager
- Compliance Operator
- ACS integration
- logging stack and forwarding

On VMs

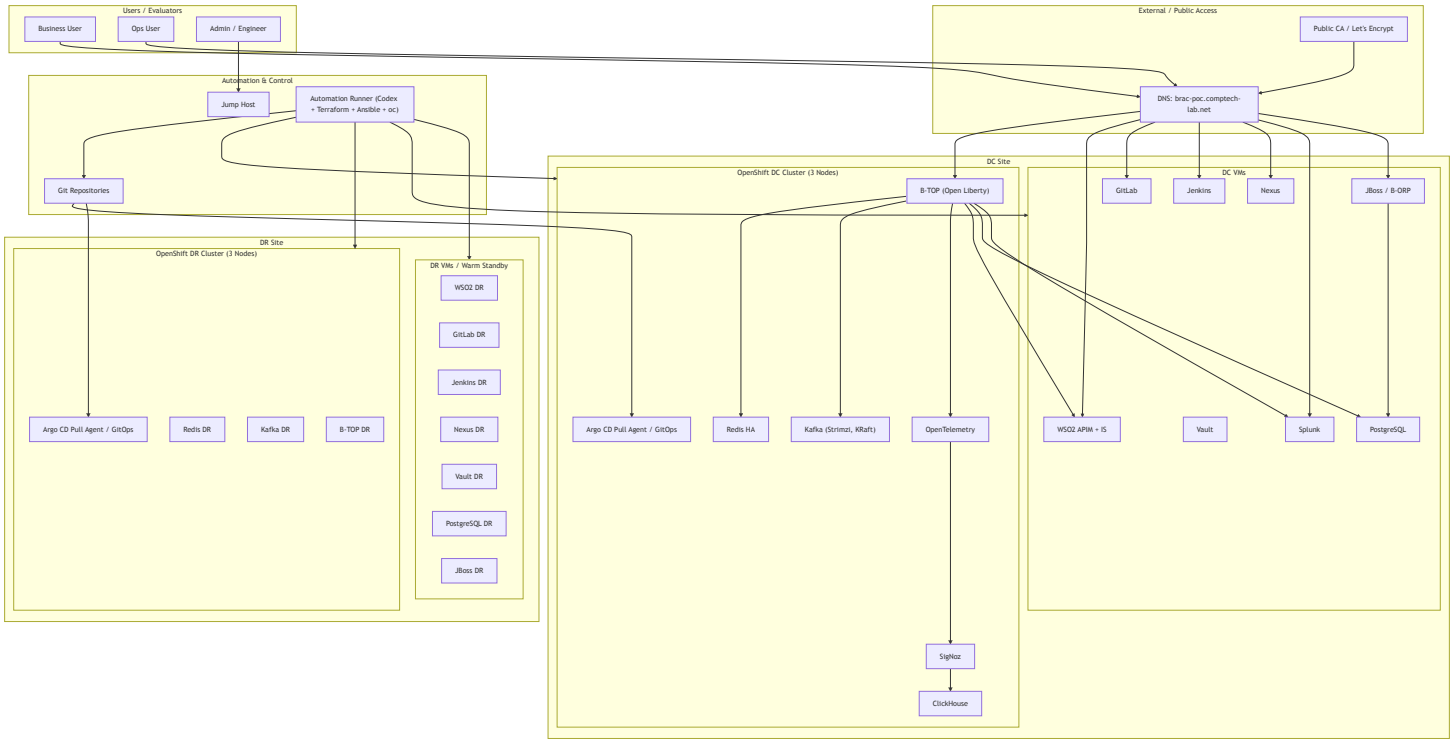
- WSO2 APIM
- WSO2 IS
- GitLab
- Jenkins
- Nexus
- Vault
- Splunk
- PostgreSQL
- JBoss / B-ORP
- automation runner
- jump host

4.2 Environment model

- all components run on Comptech internal KVM-backed infrastructure
- all configurations are stored in Git

- Codex is used from a controlled automation runner to generate, update, validate, and execute implementation assets
- OpenShift workloads are delivered using GitOps
- VM workloads are delivered using Terraform plus Ansible

5. Reference Architecture



6. Domain, DNS, and PKI Strategy

6.1 Root domain

The POC will use:

- **brac-poc.comptech-lab.net**

6.2 Cluster naming

- `api.dc.brac-poc.comptech-lab.net`
- `*.apps.dc.brac-poc.comptech-lab.net`

- `api.dr.brac-poc.comptech-lab.net`
- `*.apps.dr.brac-poc.comptech-lab.net`

6.3 Service naming

Examples:

- `btop.apps.dc.brac-poc.comptech-lab.net`
- `borp.brac-poc.comptech-lab.net`
- `apim.brac-poc.comptech-lab.net`
- `gitlab.brac-poc.comptech-lab.net`
- `jenkins.brac-poc.comptech-lab.net`
- `nexus.brac-poc.comptech-lab.net`
- `vault.brac-poc.comptech-lab.net`
- `splunk.brac-poc.comptech-lab.net`

6.4 Certificate model

Internal certificates

- Vault PKI as intermediate CA
- cert-manager integrated with Vault
- used for internal services and internal TLS

External browser-facing certificates

- cert-manager with public CA / Let's Encrypt
- used for public routes and externally viewed services

7. Banking Application Design

7.1 B-TOP – Bank Transaction Operations Platform

B-TOP is the primary Open Liberty application platform used to demonstrate transaction processing, observability, API governance, caching, messaging, and security controls.

Components

- Transaction Service
- Risk Service
- Audit Service

Functional purpose

- receive transactions
- validate requests
- publish transaction events to Kafka
- cache selected transaction state in Redis
- persist transactions and audit events
- expose APIs via WSO2
- emit logs, metrics, and traces
- support DC/DR application failover model

Key features

- end-to-end `transaction_id` propagation
- structured JSON logging
- append-only audit trail
- role-controlled access via WSO2
- rate-limited APIs
- OpenTelemetry instrumentation
- Redis integration
- Kafka integration

7.2 B-ORP – Bank Operations Reconciliation Portal

B-ORP is the JBoss-based legacy-style internal operations application.

Functional purpose

- review failed transactions
- view reconciliation summaries
- inspect exceptions
- requeue or escalate cases
- monitor batch-like operational status

Why this matters

This provides a realistic enterprise split:

- B-TOP = modern transaction platform
- B-ORP = legacy internal operations portal

8. Realism Enhancements Included in the POC

The following controls have been explicitly added to make the POC more realistic for a banking environment:

8.1 End-to-end transaction ID correlation

A transaction ID will be created or accepted at ingress and propagated through:

- WSO2
- NGINX
- B-TOP services
- Kafka headers
- Redis references
- PostgreSQL records
- logs
- traces
- Splunk search
- B-ORP views

8.2 Append-only audit trail

All critical transaction and operator actions will create immutable audit records.

8.3 RBAC

Roles will be enforced in:

- WSO2 APIs
- B-ORP actions and views

8.4 Rate limiting

WSO2 throttling policies will be applied to selected APIs.

8.5 Structured logging

Application logs will use JSON structure to support Splunk field extraction and correlation.

9. Requirement Analysis and Implementation Approach

9.1 OpenShift Platform

Requirement

Automate the provisioning of a 3-node OpenShift cluster on VMs using Terraform.

Design

Comptech will provision:

- one 3-node OpenShift cluster for DC
- one 3-node OpenShift cluster for DR

Provisioning will be done using Terraform against internal KVM-based VM infrastructure.

Implementation approach

- Terraform modules for VM creation
- cluster-specific variable files for DC and DR
- DNS and load balancer integration
- cluster install automation
- repeatable infrastructure-as-code execution from automation runner

Evidence

- Terraform code
- plan and apply output
- healthy cluster operators

- node inventory

Requirement

Configure OpenShift Data Foundation with block and object storage classes.

Design

Storage classes will be provided for:

- block-backed PVC usage
- object-backed archive use cases

These will be used for:

- application persistence where appropriate
- ClickHouse archival destination
- additional object-backed demonstrations as required

Evidence

- storage class output
- PVC binding
- archive target configuration

Requirement

Use Compliance Operator to scan against PCI-DSS standards and generate automated remediation report.

Design

Compliance Operator will be installed in the OpenShift environment and a PCI-DSS profile scan will be executed.

Evidence

- compliance scan output
- remediation report

- findings summary

Requirement

Implement ACS policy to block deployment of any image containing critical vulnerabilities.

Design

An ACS policy will be configured to deny deployment of images that exceed the allowed vulnerability threshold.

Evidence

- policy configuration
- attempted deployment of intentionally vulnerable test image
- deployment denial evidence

Requirement

Configure custom alerts within native Prometheus/Grafana stack and forward system, infrastructure, and audit logs to external mock destination.

Design

- custom alerts will be created for selected operational scenarios
- OpenShift logging will forward required logs to **Splunk** hosted externally on a VM

Evidence

- alert rule definitions
- triggered alert output
- received logs visible in Splunk

9.2 Logging and Observability

Requirement

Deploy a sample microservice instrumented with OpenTelemetry SDK and without SDK.

Design

- one B-TOP service will use direct SDK instrumentation
- another B-TOP service will use non-SDK instrumentation approach

Evidence

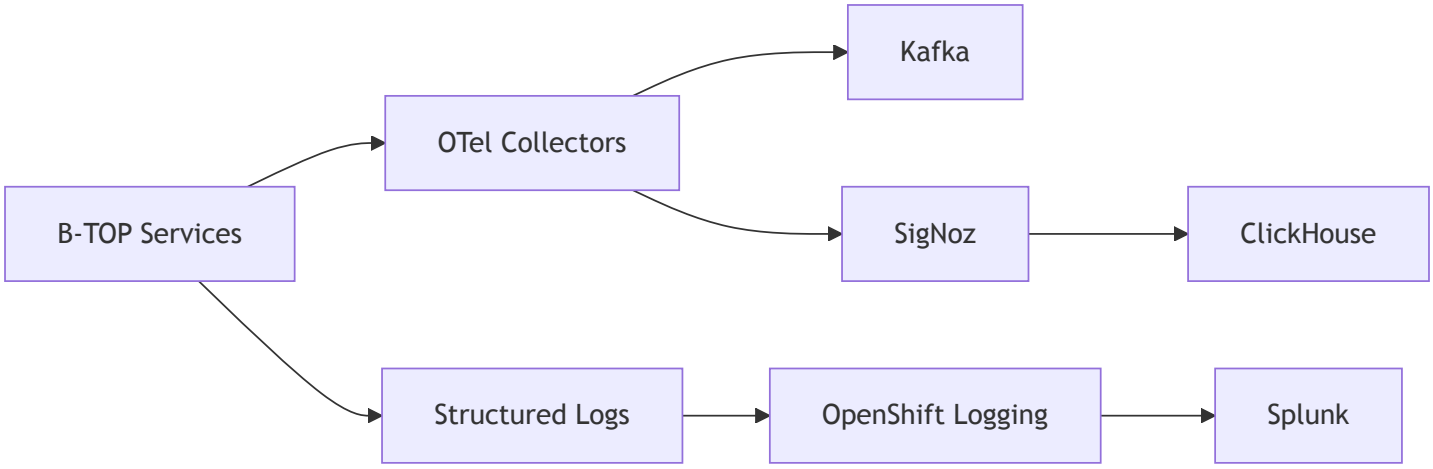
- traces visible for both services
- service map demonstrating end-to-end flow

Requirement

Build telemetry pipeline for logs, metrics, and traces using OTel, Kafka, and SigNoz.

Design

The pipeline will be:



Evidence

- trace in SigNoz
- application metrics in dashboards
- logs visible in Splunk

- collector pipeline configuration

Requirement

Implement trace sampling, log filtering, and metrics segregation.

Design

- trace sampling policies configured in collector
- log filters configured to exclude unnecessary noise
- metrics labeled and separated by service/environment

Evidence

- collector configuration
- filtered output examples
- segregated dashboards

Requirement

Configure ClickHouse with 2-day hot retention and automated cold archiving to external object storage.

Design

- ClickHouse stores hot observability data
- retention controls set to 2 days
- archive flow targets object storage

Evidence

- retention settings
- archive job configuration
- archived dataset proof

Requirement

Create dashboards for application, runtime, system, and tracing.

Design

Dashboards will include:

- HTTP request duration and count
- active requests
- DB connection usage
- JVM heap/thread metrics
- CPU and memory
- tracing dependency map and latency

Evidence

- saved dashboards
- live demo screenshots

9.3 Identity and API Management – WSO2

Requirement

Deploy distributed WSO2 APIM setup, integrate WSO2 IS, configure SSO, HA, external production-grade DB, security hardening, centralized monitoring/logging.

Design

WSO2 will run on VMs with:

- control-plane and data-plane separation
- WSO2 IS as Identity Key Manager
- PostgreSQL as external database
- SAML and OIDC flows
- basic federation demonstration
- centralized logs and monitoring integration

Evidence

- published API

- successful token-based API invocation
- SSO flow
- DB configuration
- hardened deployment notes

9.4 CI/CD and DevOps Tooling

Requirement

GitLab backup/restore, HA, DR replication; Jenkins HA and DR; conditional monorepo builds; hybrid CD; Argo CD GitOps; Nexus with external DB, backup, HA, and multi-format repositories.

Design

GitLab

- VM-based install
- backup/restore runbook
- DR approach documented and demonstrated at POC level

Jenkins

- VM-based controller
- pipeline code in Git
- monorepo conditional logic based on changed paths

Argo CD

- OpenShift GitOps only
- agent-based pull model
- Application / ApplicationSet model for cluster add-ons and app workloads

Nexus

- VM-based
- external DB
- repository formats:
  - Maven
  - npm
  - PyPI
  - Docker

Evidence

- backup/restore procedure
- successful conditional Jenkins run
- Argo sync output
- artifact push and pull from Nexus

9.5 Trivy

Requirement

Show scanning report in one dashboard for SCA and SBOM generation.

Design

Trivy will be integrated into CI to generate:

- vulnerability scan output
- SBOM artifacts
- centralized presentation layer for review

Evidence

- SBOM file
- vulnerability report
- dashboard or consolidated report view

9.6 Redis

Requirement

Deploy Redis HA cluster using OpenShift Platform, validate failover and client reconnect.

Design

Redis will run on OpenShift and provide:

- primary/replica HA topology
- failover demonstration

- B-TOP reconnect proof

Evidence

- Redis topology
- replica promotion
- successful reconnect by B-TOP or mock script

9.7 Kafka

Requirement

Provision 3-node Kafka cluster using KRaft, integrate Schema Registry, reject invalid schema produce request, configure Kafka Connect and DLQ.

Design

Kafka will run on OpenShift using **Strimzi** in **KRaft mode**, with:

- Schema Registry
- Kafka Connect
- DLQ flow
- invalid-message test path

Evidence

- broker topology
- rejected invalid produce
- connector failure routed to DLQ
- DLQ message visible

9.8 Open Liberty + NGINX

Requirement

Deploy two Open Liberty instances behind NGINX, configure header-based canary, expose health and metrics, visualize JVM Thread Count and Liveness, demonstrate HSTS, custom error pages, and rate limiting.

Design

B-TOP will be exposed through NGINX with:

- stable path
- beta path based on request header
- health and metrics exposure
- hardening controls

Evidence

- curl-based canary proof
- metrics and liveness views
- rate limiting demonstration
- security header verification

9.9 JBoss

Requirement

Set up JBoss EAP Domain Controller with at least one Host Controller and two Server Groups.

Design

JBoss will run on VMs and host B-ORP:

- one Domain Controller
- one Host Controller
- two Server Groups
- group-specific configuration demonstration

Evidence

- domain topology
- deployed WAR
- server-group-specific behavior

10. GitOps, Automation, and Codex Operating Model

10.1 Source of truth

All configurations, code, templates, manifests, and automation scripts will be kept in Git repositories.

10.2 Execution model

Terraform

Used for:

- VM provisioning
- OpenShift cluster infrastructure
- network and compute automation

Ansible

Used for:

- VM product installation and configuration
- WSO2
- GitLab
- Jenkins
- Nexus
- Vault
- Splunk
- PostgreSQL
- JBoss

Argo CD

Used for:

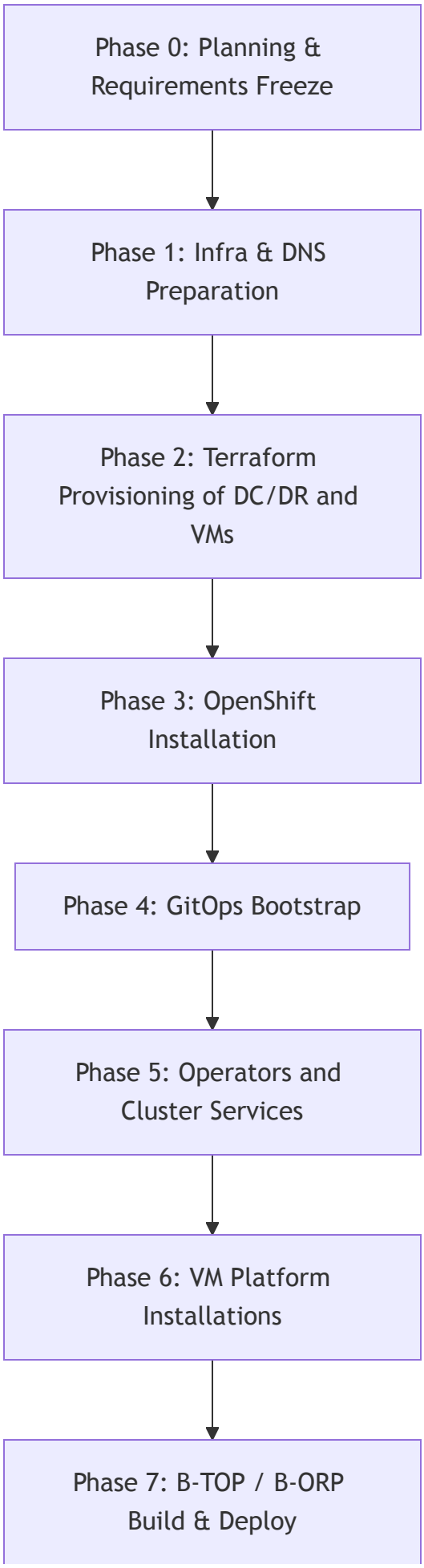
- OpenShift operator lifecycle after bootstrap
- Strimzi and Kafka resources
- Redis deployment
- B-TOP deployment
- observability components
- selected cluster add-ons

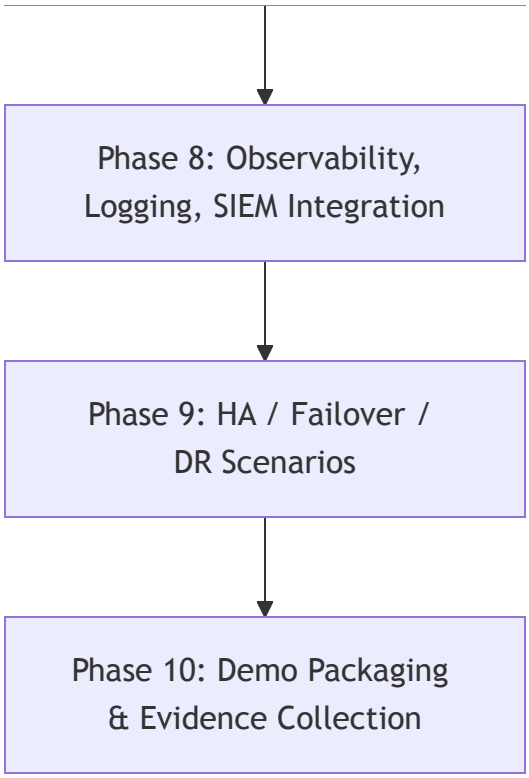
Codex

Used from a controlled automation runner to:

- update repo content
- generate implementation assets
- execute approved CLI workflows
- run validation scripts
- maintain documentation consistency

11. Deployment Sequence


Phase 0: Planning & Requirements Freeze
Phase 1: Infra & DNS Preparation
Phase 2: Terraform Provisioning of DC/DR and VMs
Phase 3: OpenShift Installation
Phase 4: GitOps Bootstrap
Phase 5: Operators and Cluster Services
Phase 6: VM Platform Installations
Phase 7: B-TOP / B-ORP Build & Deploy



12. Phased Delivery Plan

Phase 0 – Planning and Freeze

Activities

- finalize architecture
- finalize DNS and naming
- finalize IP plans
- finalize acceptance matrix
- finalize repo structure and automation model

Deliverables

- approved architecture
- acceptance criteria
- execution backlog

Phase 1 – Infrastructure Preparation

Activities

- create jump host and automation runner
- prepare KVM inventory
- prepare DNS and IP allocations
- prepare storage allocation
- prepare secrets handling and Vault bootstrap approach

Deliverables

- reachable management environment
- prepared infra variable files

Phase 2 – Terraform Provisioning

Activities

- provision DC OpenShift VMs
- provision DR OpenShift VMs
- provision supporting VM estate:
  - WSO2
  - GitLab
  - Jenkins
  - Nexus
  - Vault
  - Splunk
  - PostgreSQL
  - JBoss

Deliverables

- running VM inventory
- Terraform repo and state
- provisioning outputs

Phase 3 – OpenShift Deployment

Activities

- install DC OpenShift
- install DR OpenShift
- configure ingress and DNS
- validate cluster health

Deliverables

- 2 healthy 3-node clusters
- access and operator readiness

Phase 4 – GitOps Bootstrap

Activities

- install OpenShift GitOps
- configure Argo CD
- establish agent-based pull model
- create app-of-apps / ApplicationSet structure
- register DC and DR pull model configuration

Deliverables

- GitOps control plane active
- repo linked to clusters

Phase 5 – Cluster Services and Operators

Activities

- cert-manager
- Compliance Operator
- ACS components
- Strimzi

- Redis resources
- logging and observability operators as needed
- internal PKI integration via Vault issuer

Deliverables

- operator subscriptions
- cluster add-ons running
- cluster policies applied

Phase 6 – VM Platform Installations

Activities

- deploy Vault
- deploy PostgreSQL
- deploy WSO2 APIM + IS
- deploy GitLab
- deploy Jenkins
- deploy Nexus
- deploy Splunk
- deploy JBoss domain

Deliverables

- platform VMs functional
- Ansible playbooks committed
- validation scripts

Phase 7 – Application Implementation and Deployment

Activities

- build B-TOP services
- build B-ORP
- configure DB schemas
- configure Redis integration

- configure Kafka topics and flows
- expose B-TOP through WSO2 and NGINX
- deploy apps to DC and DR as per target model

Deliverables

- working B-TOP
- working B-ORP
- integration paths active

Phase 8 – Observability and Security Controls

Activities

- OTel collectors
- SigNoz dashboards
- ClickHouse retention and archive
- Splunk log forwarding
- Trivy scans and SBOM
- WSO2 RBAC and throttling
- transaction ID correlation
- audit trail activation

Deliverables

- dashboards
- SIEM visibility
- structured logs
- traceability proof

Phase 9 – HA, Failover, and DR

Activities

- Redis hard failover
- Kafka schema rejection and DLQ
- ACS image denial

- compliance scan
- DR application cutover scenario
- backup/restore demonstrations for selected VM products

Deliverables

- HA proof
- DR runbook validation
- test evidence pack

Phase 10 – Demo Packaging

Activities

- consolidate screenshots
- produce execution transcript
- finalize demo flow
- create quick reference commands
- produce submission artefacts

Deliverables

- final demo deck
- evidence package
- operational runbooks

13. DR Strategy

13.1 Overall model

The POC will use:

- **hot standby for the application path**
- **warm standby for supporting platforms**

13.2 Hot application path

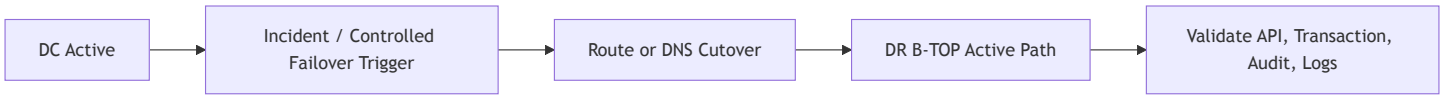
- B-TOP deployed in both DC and DR
- ingress / routing ready for cutover
- DR app stack pre-positioned

13.3 Warm platform posture

- WSO2
- GitLab
- Jenkins
- Nexus
- Vault
- JBoss
- PostgreSQL DR plan
- observability recovery posture

13.4 Failover mechanism

- controlled DNS, route, or upstream cutover
- documented activation sequence
- validation using transaction submission and lookup



14. Security and Secret Management

Vault will be used for:

- secret storage
- application credentials
- platform credentials
- internal PKI
- cert-manager integration for internal cert issuance

Secrets will not be stored in Git.

Git will contain:

- templates
- secret references
- variable examples
- policy files
- automation scripts

15. Submission Deliverables

Comptech will prepare the following deliverables for the POC:

Design deliverables

- requirement traceability matrix
- solution architecture document
- network and DNS design
- PKI and secret management design
- DR design and failover runbook

Implementation deliverables

- Terraform modules
- Ansible playbooks
- GitOps manifests
- B-TOP source code
- B-ORP source code
- Jenkins pipelines
- validation scripts

Operational deliverables

- backup and restore procedures
- HA and DR validation steps
- deployment runbook
- evidence collection package

Demo deliverables

- demo script
- dashboard views
- CLI command outputs
- screenshots and test logs

16. Assumptions

This plan assumes:

- Comptech will use its own internal infrastructure for the POC
- required DNS, IPs, and certificates can be created in the lab domain
- product licenses and trial/evaluation paths required for the POC are available where needed
- external internet access is available where public certificate issuance is required
- BRAC Bank accepts a POC that demonstrates implementation quality and architectural soundness in a controlled vendor lab environment, as allowed by the mail

17. Conclusion

This POC is designed to be:

- technically complete
- operationally realistic
- aligned to BRAC Bank's stated evaluation points
- demonstrable through clear evidence
- repeatable through automation
- secure by design
- suitable for Codex-assisted implementation without losing engineering control

The resulting environment will show not only that the requested tools can be installed, but that they can be integrated into a coherent, auditable, resilient banking platform.